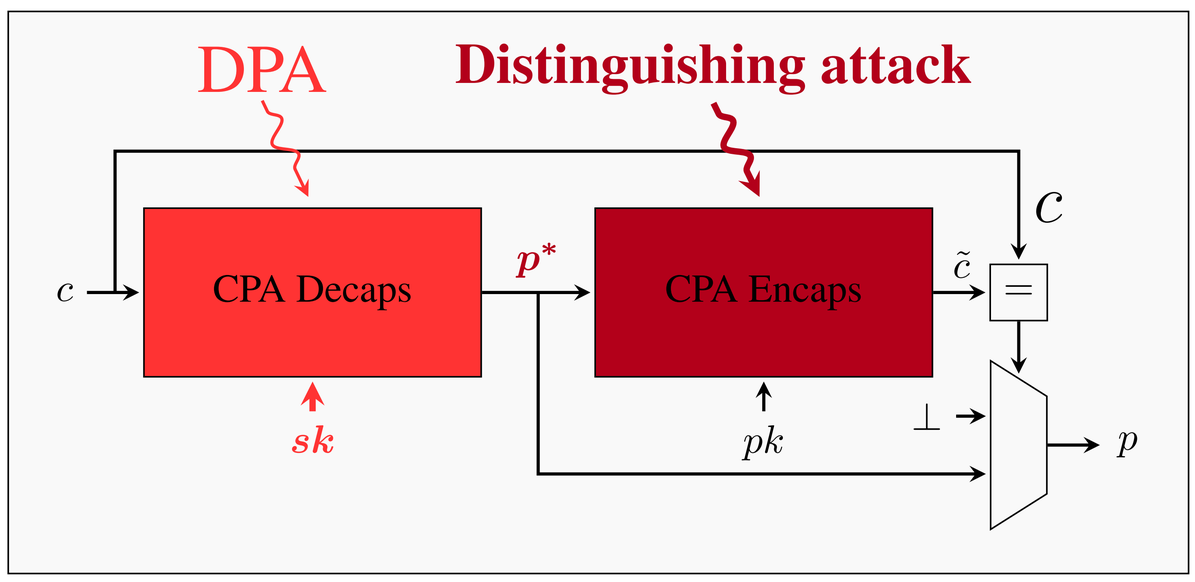
POLKA: Designing Post-Quantum PKE for Practical Leakage Resilience

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This poster summarizes the work presented in [1] and [2]

Curse of re-encryption / FOcalyspe



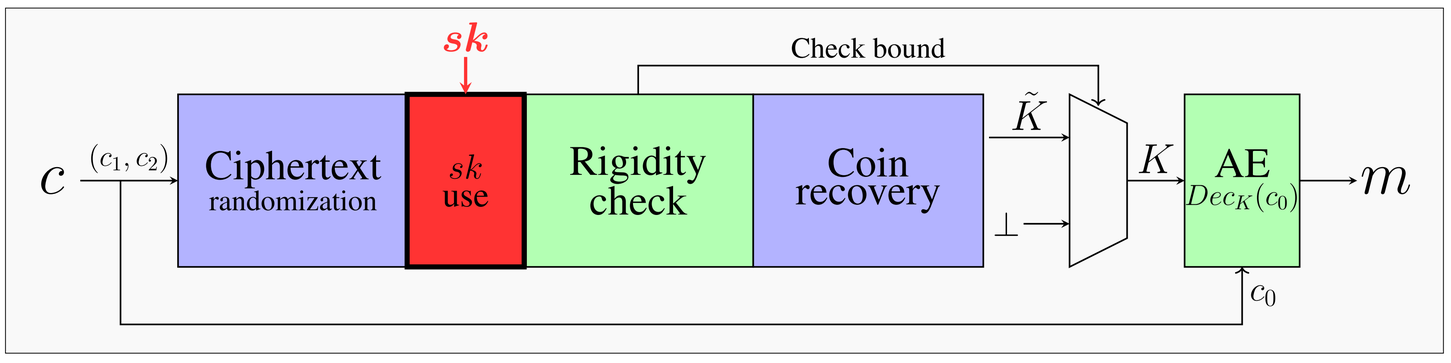
Fujisaki-Okamoto transform: CCA security from CPA KEM [3]

Take-away message: *Without designing for side-channel resistance from the outset, achieving an efficient and secure implementation is unlikely.*

FO idea. A CCA adversary can construct a malicious ciphertext c_m such that $p^* = \texttt{Decaps}(c_m)$ is equal to a single target bit of the secret key sk. Re-encryption (using the same randomness encoded in the ciphertext) verifies the validity of the ciphertext. If the ciphertext is malicious, the re-encryption outputs a random value, ensuring that a *black-box* adversary learns nothing about the sensitive value p^*.

FO and leakage. A side-channel adversary may observe leakage during re-encryption. Even distinguishing whether the input to this operation is 0 or 1 constitutes a significant attack, which would require impractical levels of masking to fully mitigate.

POLKA: CCA-secure PKE in ROM and QROM, tailored for efficient secure implementation



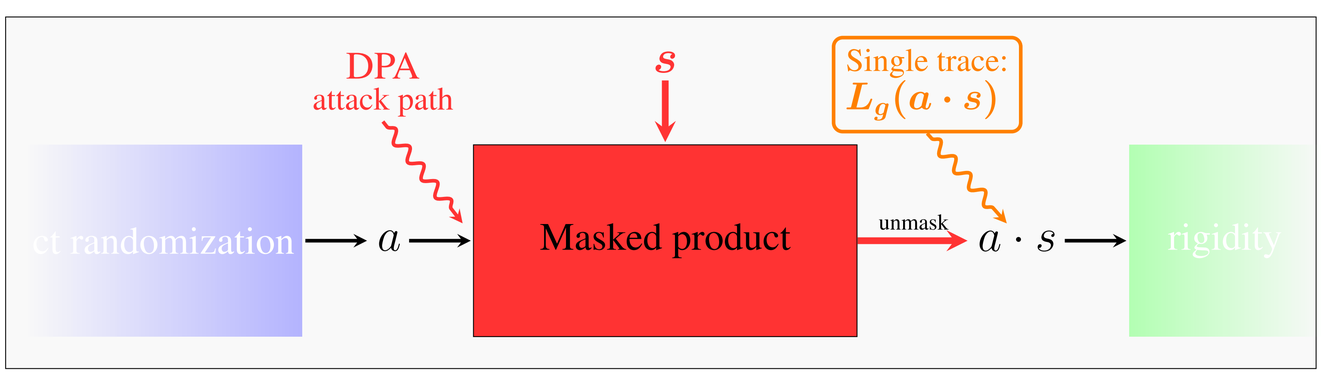
POLKA decryption step. Colors illustrates side-channel friendly design tweaks

POLKA relies on design tweaks:

- **No FO-transform:** CCA security is provided by the *rigidity property* [4]
- **Randomized ciphertext:** Leveraging RLWE ciphertexts additive homomorphy to *randomize intermediate values*, thus removing adversary control on ciphertexts.
- **Learning With Physical Rounding (LWPR) assumption:** Allows unmasking after a masked product.

The tweaks are part of the design, not an additive security layer, and therefore appear in the black-box security proofs.

Zoom on the LWPR assumption

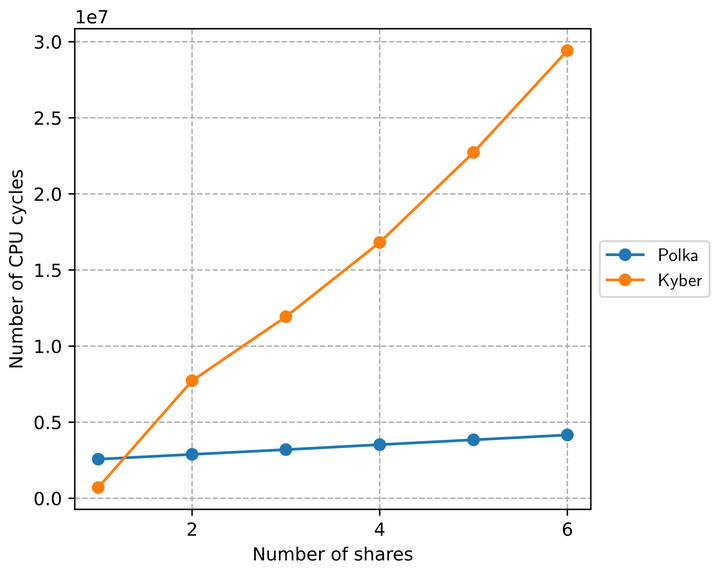


Learning With Physical Rounding:

Given a public $\mathbf{a}$ and a secret $\mathbf{s}$,
it is hard to retrieve $\mathbf{s}$ from $(\mathbf{a}, L(\mathbf{a} \cdot \mathbf{s}))$ where L is a leakage function.

- **Leakage function choices:** Definition dependency on *leakage modeling* (e.g., Hamming weight, linear), inclusion of *measurement noise* or not, and the *level of parallelism* employed.
- **Security analysis:** Current state relies on *heuristic hardness analysis* for various leakage models [5, 6], with ongoing research aiming for a *formal reduction* to the LWE problem.

POLKA masked implementation and comparison



Comparison of POLKA's and Kyber's masked decryptions

- *Public C implementation* on Github.
- *Conservative* comparison for equal number of shares.
- Key generation is not included in the comparison.
- Implementation subtleties regarding LWPR discussed in the paper. [2]

Many research directions

LWPR

- (Lattice?) reduction
- Cryptanalysis work needed

Implementation

- Analysis of concrete security level
- Efficient modular inversion for KeyGen

References

[1] Clément Hoffmann, Benoît Libert, Charles Momin, Thomas Peters, and François-Xavier Standaert, "POLKA: Towards Leakage-Resistant Post-quantum CCA-Secure Public Key Encryption".
[2] Thibaud Schoenauen, Clément Hoffmann, Charles Momin, Thomas Peters, and François-Xavier Standaert, "Leveled Software Implementation of Polka and Comparison with Uniformly Masked Kyber".
[3] Eiichiro Fujisaki and Tatsuaki Okamoto, "Secure Integration of Asymmetric and Symmetric Encryption Schemes".
[4] Daniel J. Bernstein and Edoardo Persichetti, "Towards KEM Unification".
[5] Sébastien Duval, Pierrick Méaux, Charles Momin, and François-Xavier Standaert, "Exploring Crypto-Physical Dark Matter and Learning with Physical Rounding".
[6] Clément Hoffmann, Pierrick Méaux, Charles Momin, Yann Rotella, François-Xavier Standaert, and Balazs Udvarhelyi, "Learning with Physical Rounding for Linear and Quadratic Leakage Functions".